

Insulin Negatively Impacts Glioblastoma Patient Survival

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Additional Information

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Data availability: All data were analyzed using R statistical software. A code repository for all scripts used to generate figures in this publication can be found at this link:

<https://github.com/roberthedwardsIV/GBM-Diabetes-Analysis-Figures-Tables>

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Abstract

Background: The prevalence of type 2 diabetes mellitus (T2DM) is increasing in the general population but its influence on glioblastoma patient survival is unclear.

Methods: We retrospectively analyzed the Cancer Center Registry at Brown University Health and found glioblastoma patients with (n=142) and without (n=725) T2DM who presented from 2005-2022.

Results: Patients with T2DM were older, with lower neurological performance and a lower rate of resection compared to those without. Median overall survival (OS) for diabetics was 7.9 compared to 12.3 months in non-diabetics ($p=0.0038$). Age and extent of tumor resection were significant prognostic factors. Diabetics who used insulin had a significantly shorter median OS than non-diabetics ($p=0.0003$) or diabetics who did not use insulin ($p=0.0106$). Dose matters for patients who used insulin, with a median OS of 5.2 versus 10.5 months for those who used a daily dose of >10 and <10 units, respectively ($p=0.0006$). There was no correlation between tumor size and daily (Pearson $r^2=0.0025$, $p=0.6846$) or cumulative (Pearson $r^2=0.0011$, $p=0.7859$) insulin use.

Conclusions: Glioblastoma patients with T2DM using insulin had significantly compromised survival compared to non-diabetics or non-insulin-using diabetics. Attention to diabetes and insulin use will be important in the overall management of the glioblastoma population.

Introduction

Glioblastoma is the most prevalent and aggressive primary malignant brain tumor in adults ¹. Despite significant research endeavors over past decades ², prognosis remains poor with a 5-year survival rate below 10% ³. Glioblastoma survival may be influenced by multiple medical co-morbidities including type 2 diabetes mellitus (T2DM) ^{4,5}. It has been shown that the prevalence of T2DM has been rapidly increasing and it has become a major worldwide public health problem ⁶. Published research indicates that it is also linked to a higher risk of several malignancies including renal, pancreatic, cervical and uterine cancers ⁷. However, there is conflicting evidence for an association with glioblastoma. One study revealed that T2DM was causally linked with an increased risk ⁴, while others have found no association ^{8,9}. Remarkably, one meta-analysis suggested a decreased risk of gliomas in patients with T2DM ¹⁰.

As the prevalence of T2DM rises, it is critical to determine the survival impact of this metabolic disorder in glioblastoma patients. T2DM is related to aging, and the prevalence increases from 3.0% among those at age 18 to 44, to 14.5% in those 45 to 64, and 24.4% in those 65 or older ¹¹. Furthermore, the global prevalence of prediabetes, defined as impaired glucose tolerance or abnormal fasting glucose level, was 9.1% in 2021, and it is projected to increase to 10.0% by 2045 ¹². Therefore, T2DM and prediabetes are prevalent medical co-morbidities in the glioblastoma population. Indeed, about 11 to 26% of these patients have concurrent T2DM, and the use of dexamethasone to treat cerebral edema may further trigger or exacerbate existing hyperglycemia ^{13–15}. Here, we performed a retrospective analysis of the effect of T2DM and insulin on the glioblastoma population at Brown University Health.

Methods

Data collection

Between January 2005 and December 2022, 867 glioblastoma patients received treatment at Brown University Health Cancer Institute or Rhode Island Hospital, including those initially diagnosed at a different facility or eventually transferred elsewhere. General analysis of medical comorbidities on their survival has been published ⁵. Here, we specifically focused on the effect of T2DM on the entire cohort. Of the 867 patients identified, 142 had concurrent diabetes based on their ICD-9 or -10 codes from notes on or before the date of glioblastoma diagnosis (Table 1). All identified patients with glioblastoma and diabetes were included in this analysis.

Diagnosis date, death or censoring date, patient demographics, and clinical data, including age, Karnofsky Performance Status (KPS), *O⁶-Methylguanine-DNA methyltransferase (MGMT)* promoter methylation, isocitrate dehydrogenase (IDH) mutation status, dates and types of treatment, and extent of tumor resection were tabulated from the electronic health record. When not available in physician notes, the KPS at the time of diagnosis was estimated based on the clinical description of the patient. Medical co-morbidities were documented according to ICD-9 or -10 codes from notes on or before the date of diagnosis.

Overall survival (OS) was calculated from diagnosis to death, rounded to the nearest whole number. Patients who could not be confirmed as deceased were censored at their last date of contact.

Statistical Analysis

Statistical analyses were performed in R version 4.4.1. For each co-morbidity and treatment analyzed, survival outcomes over time were determined using a Kaplan-Meier estimator (R packages: *survival*, *survminer*). Kruskal-Wallis and pairwise Wilcoxon Rank Sum tests were performed to assess differences among groups (R functions: *kruskal.test*,

pairwise.wilcox.test), and chi-squared statistics (R function: *chisq.test*) were performed on categorical variables between groups. Statistical significance was defined at $p \leq 0.05$.

For each correlation analyzed (age, tumor size, FLAIR hyperintensity, average tumor growth velocity, and average FLAIR growth velocity versus survival and average or cumulative insulin dose), Pearson correlation coefficients and correlation significance were obtained (R function: *cor(pearson)*). Lines of best fit (including r^2 and slope values) were also obtained (R function: *abline*).

To identify the insulin dose cutoff associated with differences in survival among patients taking insulin, the patient population was dichotomized using an iterative processing algorithm based on ascending insulin doses. The Python *lifelines* survival analysis library¹⁶ was used to generate a Kaplan-Meier estimator to compare the survival difference between low and high insulin groups. The low insulin group was defined as patients using an insulin dose < the cutoff, and the high insulin group was defined as patients utilizing an insulin dose $\geq$ the cutoff, at each insulin dose. The p value associated with this survival difference was then estimated.

Glioblastoma growth velocity measurements were performed according to the Response Assessment in Neuro-Oncology (RANO) criteria¹⁷. This is based on a 2-dimensional measurement of tumor size by determining the product of the 2 greatest perpendicular diameters of the tumor, using the post-gadolinium-enhanced T1-weighted Magnetization-Prepared Rapid Gradient Echo (MP RAGE) sequence. To further account for the broader infiltration of tumor cells beyond areas of enhancement, hyperintensity as seen on Fluid Attenuated Inversion Recovery (FLAIR) was also measured. Pearson correlation statistics were performed to determine an association between insulin dosing and tumor or FLAIR growth velocity.

Other notable R graphics packages used included *ggplot*, *ggforest*, *ggscatter*, *ggsurvplot*, *patchwork*, and *kableExtra*.

Results

There are some demographic differences between our diabetic (n=142) and non-diabetic (n=725) glioblastoma populations (Table 1 and Figure 1). Compared to non-diabetics, the median age of diabetics was older, 70 (range 39-93) years compared to 64 (range <1-93) years (Wilcoxon rank sum $p < 0.0001$). This is due to a greater proportion of diabetics identified at age 61 to 80 and >80 years (Chi-squared $p = 0.0008$). Furthermore, they had a lower median KPS compared to non-diabetics (Chi-squared $p = 0.0053$) and a higher percentage of them did not undergo resection, 40% compared to 32%, respectively, but this difference did not reach statistical significance (Chi-squared $p = 0.2058$). There is no difference in the number of males and females (Chi-squared $p = 0.4315$), treatment(s) received (Chi-squared $p = 0.2597$), *MGMT* promoter methylation status (Chi-squared $p = 0.1943$), IDH status (Chi-squared 0.2316), and categorical Ki-67 index (Chi-squared $p = 0.1283$). Together, these findings support the notion that diabetes is a disease of aging, and glioblastoma patients with diabetes had poorer performance status and probably fewer of them underwent aggressive resections because of this.

Table 1:

	Non-Diabetic N=725 (84%)	Diabetic N=142 (16%)	p		Non-Diabetic N=725 (84%)	Diabetic N=142 (16%)	p
Age (years)				Treatment Status			
Median	64	70	<0.0001	None	119 (17%)	31 (22%)	0.2597
Range	<1-93	39-93		RT Only	24 (4%)	7 (5%)	
<40 years	39 (5%)	1 (1%)	0.0008	Chemo Only	7 (1%)	1 (1%)	
40-60 years	239 (33%)	31 (22%)		TMZ/RT Only	159 (22%)	35 (25%)	
61-80 years	383 (53%)	89 (63%)		TMZ/RT & 1+ Line	416 (58%)	68 (48%)	
>80 years	64 (9%)	21 (15%)		MGMT Status			
Sex				Unmethylated	181 (25%)	31 (22%)	0.1943
M:F	1.4	1.2	0.4315	Methylated	236 (33%)	39 (28%)	
Males	417 (58%)	76 (54%)		Unknown	308 (43%)	72 (51%)	
Females	308 (43%)	66 (47%)		IDH Type			
Baseline KPS				Wild-Type	429 (60%)	79 (56%)	0.2316
Median	80	70	0.0053	Mutant	10 (2%)	0 (0%)	
Range	0-100	0-100		Unknown	286 (40%)	63 (45%)	
0-60	220 (31%)	59 (42%)		Ki-67 Index			
70-80	247 (35%)	51 (36%)		0-20%	80 (12%)	14 (10%)	0.1283
90-100	243 (34%)	30 (22%)		21-40%	221 (31%)	45 (32%)	
Resection Status				41-60%	148 (21%)	23 (17%)	
None	232 (32%)	56 (40%)	0.2058	61-80%	56 (8%)	4 (3%)	
STR	237 (33%)	39 (28%)		81-100%	5 (1%)	1 (1%)	
GTR	256 (36%)	47 (34%)		Unknown	215 (30%)	55 (39%)	
				Survival Analysis			
				Median OS (months)	12.3	7.9	0.0038
				95% CI	11.1 - 13.2	6.4 - 10.8	
				Events/N	602 (83%)	138 (98%)	

Table 1: Demographic data on non-diabetic and diabetic patients.

Figure 1:

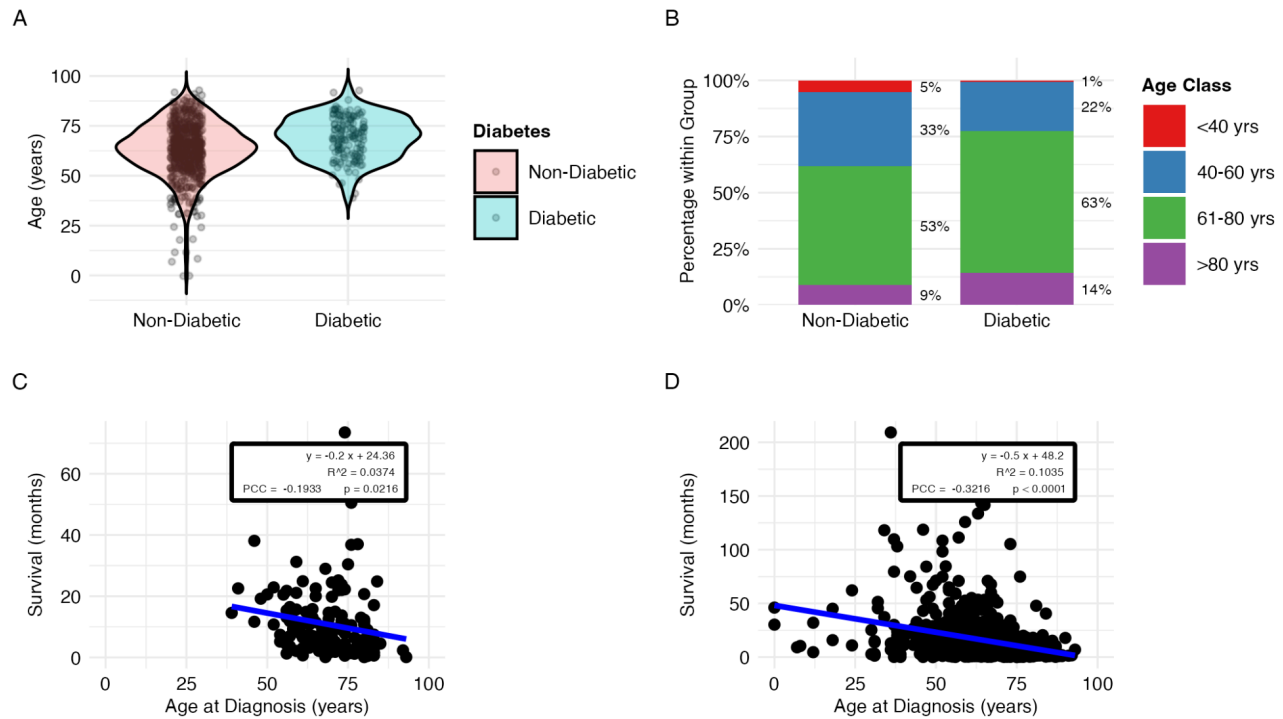


Figure 1: Age and survival in non-diabetic and diabetic patients with glioblastoma. Age distribution and median age between non-diabetics and diabetics (A). Categorical age distribution in non-diabetics according to age <40, 41-60, 61-80 and >80 years (B). Pearson correlation between age and survival in diabetics (C) and non-diabetics (D).

Glioblastoma patients with diabetes had shortened survival compared to the rest of the population

Compared to non-diabetics, glioblastoma patients with diabetes had a significantly shortened survival (Table 1). Median OS was 7.9 (95%CI 6.4-10.8) months in diabetics compared to 12.3 (95%CI 11.1-13.2) months in non-diabetics ($p=0.0038$) (Figure 2A). This prompted us to perform an in-depth analysis of the influence of diabetes on glioblastoma patient survival. At the time of data extraction, all 142 patients were dead or censored.

Figure 2:

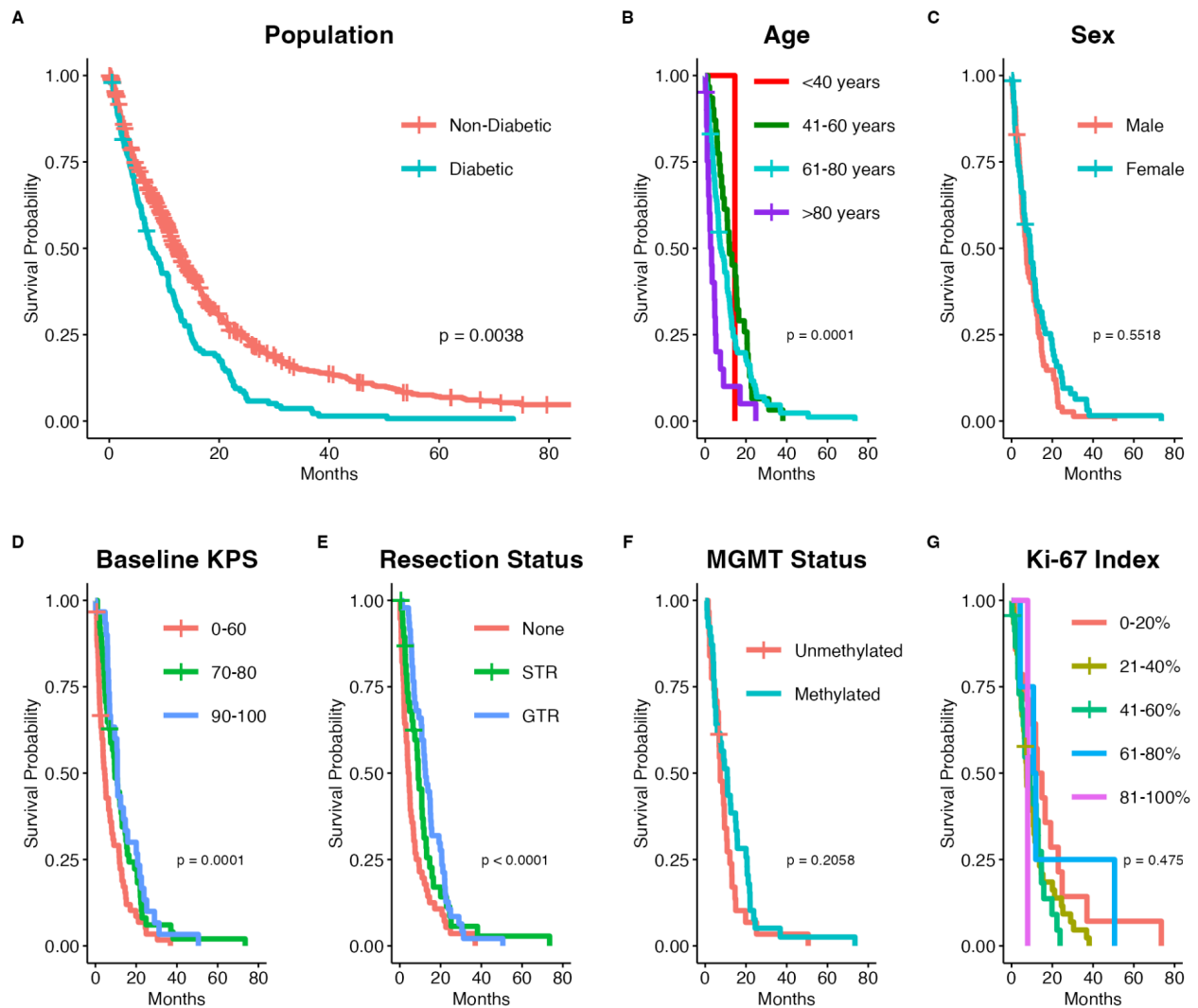


Figure 2: Kaplan-Meier survival and prognostic factors in non-diabetic and diabetic patients with glioblastoma. Non-diabetics had a longer median OS than diabetics (A). Among the diabetics, age (B), baseline KPS (D) and resection status (E) were significant prognostic factors, but not sex (C), MGMT promoter methylation status (F), or Ki-67 index (G).

Univariate and multivariate analyses revealed prognostic factors that correlate with overall survival

Our diabetic and non-diabetic cohorts have the expected prognostic factors for glioblastoma. For diabetics, univariate analyses showed expected survival correlation with age

(when stratified into increments from <40, 40-60, 61-80, and >80 years), baseline KPS, and surgical resection status, but not sex or Ki-67 (Figure 2) ⁵. Multivariate Cox proportional hazard analysis in the non-diabetic cohort showed significant correlations between survival and age, KPS, extent of surgical resection, and *MGMT* promoter methylation status (Table 2A). Among the diabetics, resection status was significant and an increase in hazard ratio was noted only in patients at age >80 years (Table 2B), suggesting that aggressive resection is probably warranted for those up to the age of 80. There was also a trend for decreased hazard ratio in those with KPS 90-100 and methylated *MGMT* (Table 2B). The IDH status was not informative in either cohort because only 10 had mutated IDH in the nondiabetic and none in the diabetic cohort (Table 1). No specific treatment had an influence on survival (Table 1 and Supplementary Figure 1B).

Table 2:

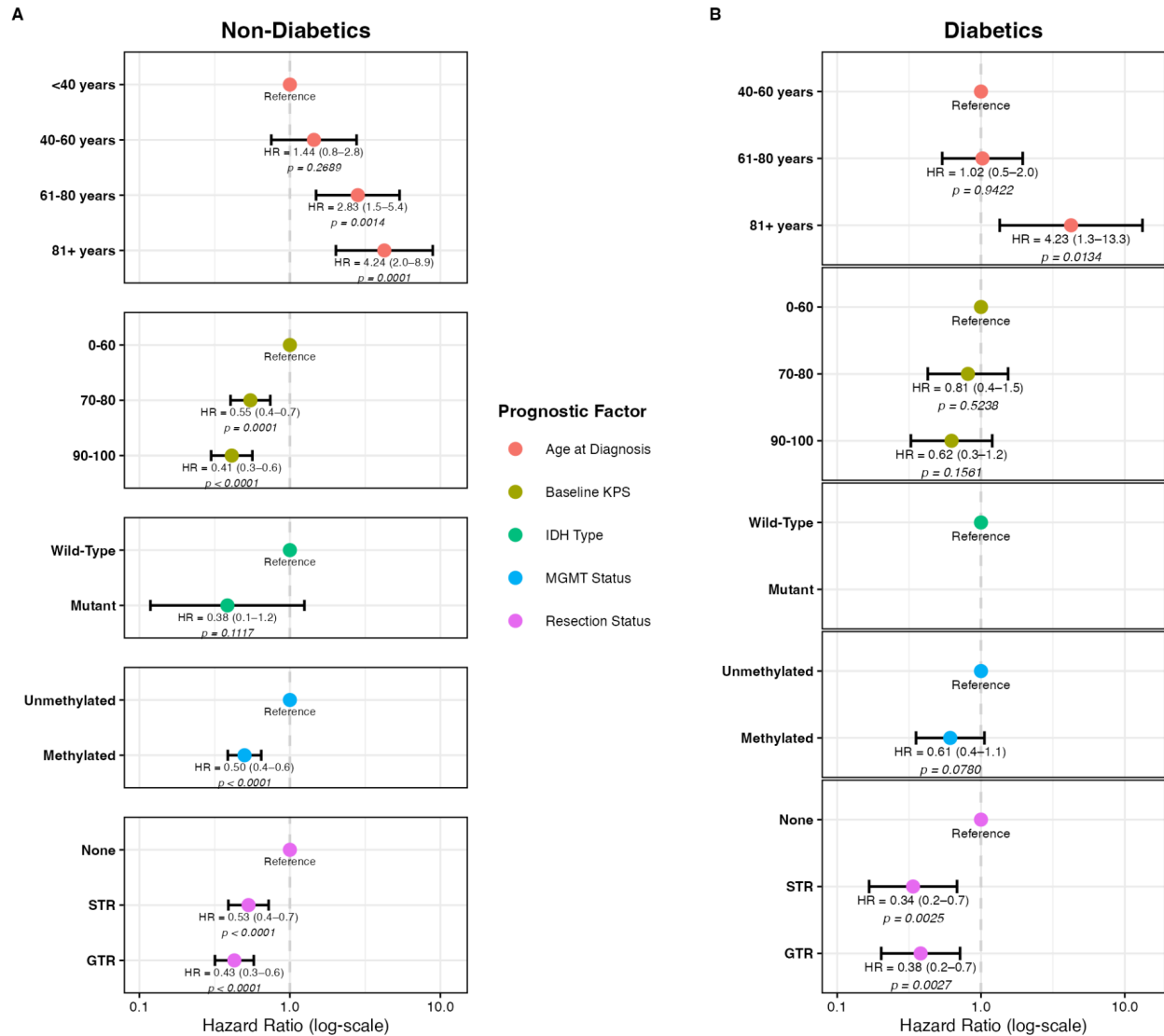


Table 2: Multivariate analysis of prognostic factors between non-diabetic (**A**) and diabetic (**B**) patients with glioblastoma.

A large proportion of patients had unknown status in *MGMT* methylation (n=72 or 51%). Therefore, we performed a sensitivity analysis to determine the magnitude of statistical variance when these 72 patients were incorporated into the unmethylated or methylated cohorts (respectively denoted here as unmethylated* or methylated*). As expected, the unmethylated*

cohort (n=31+72=103) still had a median OS near 7.4 (95%CI 6.0-10.5) months, but it is showing a trend for significance in the methylated cohort's improved survival (n=31) (p=0.1023) (Supplementary Figure 2A). Furthermore, the methylated* cohort (n=39+72=111) had a median OS of 7.9 (95%CI 5.9-11.5) months, which is similar to the unmethylated cohort (n=31) (p=0.8803) (Supplementary Figure 2B). Despite a smaller sample size in our glioblastoma patients with concurrent T2DM, this sensitivity analysis identified a trend for the prognostic significance of *MGMT* promoter methylation status^{18,19}.

We next performed a similar sensitivity analysis for IDH mutation status. All were IDH wild-type (n=79) and none were mutated, while 63 (45%) had unknown status. Therefore, we tested these 63 patients as IDH mutated (denoted as IDH mutated*) and they had a median OS of 10.8 (95%CI 5.2-13.7) months, compared to the IDH wild-type cohort that had a median OS of 7.3 (95%CI 6.0-9.5) (p=0.7041) (Supplementary Figure 1A). As expected, the difference in the median OS is statistically insignificant between the IDH mutated* and wild-type cohorts even though they are numerically different, probably because IDH mutated glioblastomas are uncommonly found in older adults. Collectively, we believe that the baseline characteristics of our glioblastoma patients with diabetes are not dissimilar to those without diabetes or the general glioblastoma population⁵.

Insulin use compromises glioblastoma survival

Insulin is a *bona fide* cellular growth factor and its cancer-promoting properties have been hypothesized previously^{20,21}. We therefore proceeded to compare the Kaplan-Meier survival and Cox proportional hazard characteristics of diabetic patients who used insulin and non-insulin medications, as well as those who did not have diabetes, and noted a significant difference among these groups (Kruskal-Wallis p=0.0005) (Figure 3A and 3B). Diabetic patients who used insulin (n=120) had a significantly compromised median OS compared to non-diabetics (n=725): 7.0 (95%CI 5.8-9.4) versus 12.3 (95%CI 11.1-13.2) months, respectively

($p=0.0003$) (Figure 3C). Furthermore, insulin-using diabetics ($n=120$) also had a significantly reduced median OS compared to non-insulin-using diabetics ($n=22$): 7.0 (95% CI 5.8-9.4) versus 15.4 (95%CI 9.6-25.2) ($p=0.0106$) (Figure 3D). However, there was no difference between the non-diabetics ($n=725$) and non-insulin-using diabetics ($n=22$) ($p=0.3183$) (Figure 3E).

*****INSERT FIGURE 3 HERE*****

To further explore medication effects on our diabetics, we divided them into 4 subgroups into those who used (i) insulin ($n=34$), (ii) non-insulin diabetes medication(s) ($n=10$), (iii) both ($n=86$), or (iv) neither ($n=12$), and noted a statistical difference among them (Kruskal-Wallis $p=0.0512$) (Figure 3A). The non-insulin medications include glyburide, glimepiride, glipizide, sitagliptin, linagliptin, metformin, rosiglitazone, pioglitazone, acarbose, canagliflozin, empagliflozin, ertugliflozin, colesevelam, dulaglutide, exenatide, liraglutide, or any of the combinations. Cox proportional hazard analysis showed increased hazard ratios for those who used insulin with or without concurrent non-insulin diabetes medications (Figure 4B); in contrast, those who used only non-insulin diabetes medications had a decreased hazard ratio (Figure 4B). The median OS of patients who used insulin alone was 6.6 (95%CI 3.2-10.8) months which is significantly shorter compared to those using non-insulin diabetes medication(s), 11.8 (95%CI 9.0-N/A) months ($p=0.0467$), or no diabetes medication 17.1 (95%CI 11.5-N/A) months ($p=0.0427$) (Figure 4C and 4D). There was a trend toward shortened survival among those who use both insulin and non-insulin diabetes medication(s), median OS 7.4 (95%CI 5.8-10.8), compared to those using only non-insulin medication(s), median OS 11.8 (95%CI 11.5-9.0) ($p=0.0565$) (Supplementary Figure 3A), or neither 17.1 (95%CI 11.5-N/A) ($p=0.1238$) (Supplementary Figure 3B). Together, insulin usage appears to correlate with a shortened survival in our diabetic cohort. Furthermore, no survival difference was noted between patients who used non-insulin diabetes medication(s) compared to those not using any medication at all

($p=0.7919$) (Supplementary Figure 3C), as well as patients who use insulin compared to those using both insulin and non-insulin medication(s) ($p=0.2573$) (Supplementary Figure 3D). Collectively, insulin use, either alone or in combination with other diabetes medication(s), appears to have a negative impact on glioblastoma patient survival.

INSERT FIGURE 4 HERE

Shorter survival is associated with higher insulin usage

We next explored the relationship between insulin dose and diabetic patient survival. Fifty-six (39%) patients did not have insulin dosage data in the medical records. In the remaining 86 patients, we determined the cutoffs for average daily insulin dosage and cumulative insulin use that lent itself to the greatest survival difference and the lowest p value between the two dichotomized cohorts (Figures 5). For average daily insulin dose, we found that this cutoff was near 10 units/day (Figures 5A). We then dichotomized the diabetic population based on the average daily insulin usage into (i) high-dose (>10 units/day) and (ii) low-dose (<10 units/day). The Kaplan-Meier survival statistics showed a significant difference between these 2 subgroups (Figures 5B). Importantly, patients who used high-dose insulin on a daily basis had a markedly shortened survival compared to those who used low dose: median OS 5.2 (95%CI 4.1-9.3) versus 10.5 (95%CI 7.4-13.1) months, respectively ($p=0.0006$). Similarly, the cutoff for cumulative insulin dose was near 50 units/day (Figure 5C). Patients who used high cumulative doses of insulin also had a statistically significant shortened survival compared to those who received low cumulative doses: median OS 6.7 (95%CI: 5.2-9.4) versus 11.5 (95%CI 7.4-21.8) months ($p=0.0104$) (Figure 5D). Collectively, these data indicate insulin's negative effect on glioblastoma patient survival is dose-dependent and it is inversely related to the dose administered.

INSERT FIGURE 5 HERE

Lack of association between insulin dosage and glioblastoma tumor size or growth velocity

We subsequently asked whether high insulin usage would correlate with tumor size or the growth velocity of glioblastoma. This analysis excluded an additional 51 patients because they (i) did not take insulin, (ii) did not have an initial resection, (iii) had ≤ 1 MRI scan archived at our institution, or (iv) did not have an available immediate post-resection MRI. Furthermore, to make comparisons uniform, we excluded tumor velocity measurements after a second resection. In the remaining subset of 35 patients, we found no correlation between the average daily insulin dose and tumor size as measured using the gadolinium-enhanced T1-weight MP RAGE (Pearson $r^2=0.0025$, $p=0.6846$) (Figure 6A) or FLAIR images (Pearson $r^2<0.0001$, $p=0.9548$) (Figure 6B). Similarly, there was no correlation between the cumulative insulin dose and tumor size as estimated using the gadolinium-enhanced T1-weight MP RAGE (Pearson $r^2=0.0011$, $p=0.7859$) (Figure 6C) or FLAIR images (Pearson $r^2<0.0001$, $p=0.8254$) (Figure 6D).

*****INSERT FIGURE 6 HERE*****

Glioblastoma growth velocity was computed by determining the difference in tumor size between first post-resection MRI and each of the subsequent MRI scans divided by the time interval between them. No significant Pearson correlation was noted between daily insulin dose and tumor size (Supplementary Figure 4A). Although there is a slight but significant downward sloping relationship between daily insulin dose and FLAIR images (Supplementary Figure 4B), this may be just a stochastic anomaly since the correlation was insignificant between the cumulative insulin dose and tumor size (Supplementary Figure 4C) and FLAIR images (Supplementary Figure 4D). Alternatively, the downward sloping could be from decreased inflammation as a result of normalized glucose in the interstitium.

Discussion

An important finding from our analysis is that insulin has an overriding negative survival impact on our glioblastoma patients, and this relationship is dose dependent. The survival of diabetics in our cohort who used insulin was shortened by more than one-third compared to those who did not. Furthermore, there was no survival difference between non-diabetics and diabetics who did not use insulin. Prior retrospective studies did not focus on insulin utilization, probably due to the significant fluctuations in dose administered, and the emphasis was on the presence or absence of T2DM or the severity of hyperglycemia in these patients ^{13,22}. Importantly, fast-acting insulin is the quickest and most effective way to ameliorate hyperglycemia ²³ and, when patients are placed on insulin, few are switched to non-insulin secretagogues or other drugs. However, we meticulously extracted the dosage from electronic medical records and identified the average daily dose and the cumulative dose for each patient. Our effort revealed that there is indeed a dose-dependent relationship between survival and daily insulin dosage as well as cumulative insulin use. If this is confirmed in future studies, glycemic control without using insulin or with non-insulin diabetic medications may be a judicious approach for managing T2DM in glioblastoma patients.

Derr et al. performed a retrospective analysis of hyperglycemia among newly diagnosed glioblastoma patients enrolled in 5 phase II clinical trials. They found that those with a time-weighted mean serum glucose level of >137 mg/dL had a significantly decreased survival, with a hazard ratio of 1.57, compared to those with a level of <94 mg/dL after controlling for corticosteroid use and other confounders ¹⁵. They posit that intensive management of corticosteroid-induced hyperglycemia may improve the longevity of these patients. Other studies also revealed an association with survival, at a serum glucose cutoff of 90 mg/dL ²⁴, 112 mg/dL ²⁵, 113 mg/dL ²², 167 mg/dL ²⁶ or 180 mg/dL ^{27,28}, while others did not find an association ^{29,30}. However, it is still unclear whether poor survival associated with high levels of

hyperglycemia in these patients is due to intensive insulin use, complications from T2DM, extent of dexamethasone utilization, or any of the combinations above ³¹.

A remarkable finding from our dataset is that diabetics had a shortened OS compared to non-diabetics. However, the association between T2DM and cancer patients is controversial, where some studies found an association^{28,30,32} while others did not^{8,27,29,33–37}. Most of these studies did not investigate complications associated with T2DM or their contribution to patient survival. Regardless, there are definitely more negative than positive studies, suggesting that T2DM *per se* in the glioblastoma population may not attenuate their survival.

Insulin is a *bona fide* growth factor that binds to both insulin and insulin-like growth factor receptors, triggering cellular growth and proliferation signals via the phosphoinositide-3 kinase (PI3K) pathway ^{38,39}. This pathway also regulates metabolic functions in tissues and organs for homeostasis ⁴⁰. According to The Cancer Genome Atlas, alterations in PI3K were found in 40% of glioblastoma, including amplifications as well as activating mutations and fusions ⁴⁰. However, the brain-penetrant PI3K inhibitor, GDC-0084 (buparlisib), showed no efficacy in treating recurrent glioblastoma despite achieving pharmacodynamic concentrations in the patient's brain and the tumor ^{41,42}. This is most likely due to drug-induced hyperglycemia resulting in a secondary insulin feedback escape mechanism, and suppression of this insulin feedback in mouse models enhanced the efficacy of PI3K inhibitors ^{43,44}. The collective data indicate that PI3K inhibition alone is a useless therapeutic strategy for glioblastoma without blocking the possible secondary surge of endogenous insulin as a result of hyperglycemia or suppressing the use of exogenous insulin to normalize high serum glucose levels.

This study has several limitations. First, our patients are heterogeneous but they possess known prognostic factors that are typically found in the glioblastoma population. Second, it is difficult to resolve individual effects from dexamethasone use, administration of

insulin, or hyperglycemia triggering compensatory endogenous hyperinsulinemia or administration of external insulin for lowering high serum glucose. However, our data on the association between insulin dose and survival is very compelling. Lastly, there is no correlation between insulin and tumor size. This lack of correlation is probably secondary to the infiltrative nature of glioblastoma, making precise measurement of tumor size difficult. The bi-dimensional tumor measurement, consisting of the product of two longest perpendicular diameters, has limitations in the RANO criteria ⁴⁵. Therefore, volumetric measurement of tumor size in RANO 2.0 probably provides a more accurate assessment ¹⁷, but it is still difficult to account for the microscopic glioblastoma cells that infiltrate into the adjacent brain.

In conclusion, our analysis reveals that our diabetic glioblastoma patients who used insulin had a significantly compromised survival compared to non-diabetics or non-insulin-using diabetics. Those who received higher daily or cumulative insulin doses had a shorter survival than others who used lower doses. Therefore, in addition to administering cancer drug treatments, oncologists' attention to diabetes and insulin use will be paramount in the overall management of the glioblastoma population.

Figure Legends

Figure 1: ~~Age and survival in non-diabetic and diabetic patients with glioblastoma. Age distribution and median age between non-diabetics and diabetics (A). Categorical age distribution in non-diabetics according to age <40, 41-60, 61-80 and >80 years (B). Pearson correlation between age and survival in diabetics (C) and non-diabetics (D).~~

Figure 2: ~~Kaplan-Meier survival and prognostic factors in non-diabetic and diabetic patients with glioblastoma. Non-diabetics had a longer median OS than diabetics (A). Among the diabetics, age (B), baseline KPS (D) and resection status (E) were significant prognostic factors, but not sex (C), MGMT promoter methylation status (F), or Ki-67 index (G).~~

Figure 3: ~~Insulin usage in diabetic patients with glioblastoma. Kaplan-Meier survival among non-diabetics, non-insulin-using diabetics, and insulin-using diabetics (A). Cox proportional hazard ratio of all diabetics, non-insulin-using diabetics and insulin-using diabetics (B). Insulin-using diabetics had a shortened survival compared to non-diabetics (C) and non-insulin-using diabetics (D). But there was no survival difference between non-diabetics and non-insulin-using diabetics (E).~~

Figure 4: ~~The influence of insulin versus non-insulin diabetes medications (NIDM(s)) on patient survival. Kaplan-Meier survival showed a difference among patients using insulin; NIDM(s), both and neither (A). Cox proportional hazard ratio among patients using insulin; NIDM(s), both or neither (B). Shortened survival was observed in patients using insulin compared to those using NIDM(s) (C) or neither (D).~~

Figure 5: ~~Low versus high insulin usage. Distribution of average daily insulin among diabetic patients (A) and Kaplan-Meier survival difference between low versus high daily insulin usage at the optimal cutoff of 10 units/day (B). Distribution of cumulative insulin among diabetic~~

patients (C) and Kaplan-Meier survival difference between low versus high cumulative insulin usage at the optimal cutoff of 50 units (D).

Figure 6: Glioblastoma tumor size and insulin dose. No difference was seen between average daily insulin dose and bi-dimensional tumor measurements from T1-weighted post-gadolinium MP-RAGE (A) and FLAIR (B) images. No difference was also observed between cumulative insulin dose and bi-dimensional tumor measurements from T1-weighted post-gadolinium MP-RAGE (C) and FLAIR (D) images.

Table Headings

Table 1: Demographic data on non-diabetic and diabetic patients.

Table 2: Multivariate analysis of prognostic factors between non-diabetic (A) and diabetic (B) patients with glioblastoma.

Supplementary Figure Legends (ADD SUPP FIGURES HERE)

Supplementary Figure 1: IDH status and treatment effects in diabetic patients with glioblastoma. Sensitivity analysis of IDH status revealed no Kaplan-Meier survival difference between patients with IDH wild-type and mutated* status (unknown categorized as mutated) (A). No difference in survival according to treatment effects: (i) no treatment, (ii) radiation only, (iii) chemotherapy only, (iv) concurrent chemotherapy and radiation, and (v) concurrent chemotherapy and radiation with at least 1 line of subsequent treatment (B).

Supplementary Figure 2: Sensitivity analysis of *MGMT* promoter methylated status. Sensitivity analysis revealed a trend favoring patients with *MGMT* methylated status compared to those with unmethylated* status (unknown + unmethylated) (A). No survival difference was

~~noted between patients with *MGMT* unmethylated status and those with methylated* status (unknown + methylated) (B).~~

~~Supplementary Figure 3: The influence of insulin versus non-insulin diabetes medications (NIDM(s)) on patient survival. Compared to patients using both insulin and NIDM(s), there is a trend favoring those using NIDM(s) (A) or neither (B). No difference was seen in survival between patients using NIDM(s) and neither (C), as well as those using insulin and both insulin and NIDM(s) (D).~~

~~Supplementary Figure 4: Glioblastoma tumor growth velocity and insulin dose. No difference was noted between average daily insulin dose and tumor growth velocity as measured using T1-weighted post-gadolinium MP-RAGE images (A). There is a slight but downward sloping relationship between average daily insulin dose and tumor growth velocity as measured using FLAIR images (B). No difference was seen between the cumulative insulin dose and tumor growth velocity as estimated using T1-weighted post-gadolinium MP-RAGE (C) or FLAIR images (D).~~

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